Beyond MLR Lab 1: Modeling center as a fixed effect

A multicenter rehabilitation trial

We return to the rehabilitation trial from the lecture. In this simulated study, 120 patients recovering from knee replacement receive either **standard care** or an **enhanced rehabilitation program**. Standard care consists of weekly physiotherapy and home exercises; enhanced care adds a weekly supervised session. The outcome is the distance walked during a six-minute walk test at 12 weeks, measured in meters. A larger distance indicates better walking performance.

The study includes six centers. Within each center, 10 patients are randomly assigned to standard care and 10 to enhanced care. This is a **randomized block design**, with center as the blocking factor. These are simulated teaching data, not observations from an actual clinical trial.

Our question is: **What is the estimated difference in expected walking distance between enhanced and standard care, and how precisely is it estimated?** We will explore differences between centers and fit a model that includes center as a fixed effect.

By the end of this lab, you should be able to visualize treatment and center differences, fit and interpret the fixed-effects model, assess the overall contribution of center using an analysis of variance table, and check the model assumptions.

> Working through the lab
>
> Run the code in order in an R script, and record your answers to the questions. The lab page shows the code; you generate the results yourself.

Setup and data

We use `ggplot2` for figures and `dplyr` for summaries. The following code installs any missing packages and loads them.

```
packages <- c("ggplot2", "dplyr")
missing_packages <- packages[
  !vapply(packages, requireNamespace, logical(1), quietly = TRUE)
]
if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}

library(ggplot2)
library(dplyr)
```

Download the rehabilitation dataset. Save it in a folder named `data` inside your working directory. Keep your R script in that working directory. You can use `getwd()` to check where R is looking for files.

```
rehab <- read.csv("data/rehabilitation_week1.csv")

# Specify the order of the categories explicitly.
rehab$treatment <- factor(rehab$treatment,
                          levels = c("Standard", "Enhanced"))
rehab$center <- factor(rehab$center,
                       levels = paste("Center", LETTERS[1:6]))

head(rehab)
str(rehab)
table(rehab$center, rehab$treatment)
```

Each row represents one patient. The variables are `center`, `treatment`, and `distance` (six-minute walk distance in meters). A **factor** tells R that a variable represents categories rather than a continuous measurement. The order of the factor levels will matter when we choose how to represent these categories in the model.

> Question 1
>
> Check the allocation table. How many patients are in each treatment group within each center? Why is this allocation useful when comparing the treatments?

Explore treatment and center differences

Create a plot that shows the individual walking distances and the treatment-specific means within each center. Add the overall mean for each treatment as a reference line. This lets us examine center differences while keeping the two treatment groups separate.

First calculate the mean walking distance for each treatment within each center, and then the overall mean for each treatment.

```
center_treatment_means <- rehab |>
  group_by(center, treatment) |>
  summarise(mean_distance = mean(distance), .groups = "drop")

treatment_means <- rehab |>
  group_by(treatment) |>
  summarise(mean_distance = mean(distance), .groups = "drop")
```

Use these summaries together with the individual observations to create the plot.

```
center_plot <- ggplot(rehab,
                      aes(x = center, y = distance, color = treatment)) +
  geom_hline(data = treatment_means,
             aes(yintercept = mean_distance, color = treatment),
             linetype = "dashed", linewidth = 0.7, show.legend = FALSE) +
  geom_point(position = position_jitterdodge(jitter.width = 0.18,
                                             jitter.height = 0,
                                             dodge.width = 0.64,
                                             seed = 20261104),
             alpha = 0.35, size = 1.8) +
  geom_point(data = center_treatment_means,
             aes(y = mean_distance),
             position = position_dodge(width = 0.64),
             shape = 18, size = 4, show.legend = FALSE) +
  scale_color_manual(values = c(Standard = "#173F48",
                                Enhanced = "#D58A3D")) +
  labs(x = "Center", y = "Six-minute walk distance (m)",
       color = "Treatment") +
  guides(color = guide_legend(override.aes = list(alpha = 1, size = 3))) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "top")

center_plot
```

The small points are individual observations. They are shifted slightly along the horizontal axis to make overlapping points easier to see; their walking distances are unchanged. The diamonds show the observed treatment means within each center. The dashed lines show the overall treatment means across all six centers.

> Question 2
>
> Does the plot suggest substantial differences in walking distance between centers? Support your answer using the center-specific means and the variation among patients within centers.

Choose how to code the factors

A regression model represents a categorical variable using numerical columns. The **coding scheme** determines what the intercept and factor coefficients mean. We will use two schemes in the same model: dummy coding for treatment and effects coding for center. These choices make the coefficients useful for our research question.

Dummy coding for treatment

With **dummy coding**, one category is the reference. For our two treatments, we use standard care as the reference and represent treatment by a single indicator:

Treatment	Indicator T_{ij}
Standard	0
Enhanced	1

In a model containing $\beta_0 + \beta_1 T_{ij}$, the predicted standard-care outcome is β_0, and the predicted enhanced-care outcome is $\beta_0 + \beta_1$. The treatment coefficient therefore directly answers our question: it is the enhanced-minus-standard difference.

```
# The first factor level, Standard, is the reference.
contrasts(rehab$treatment) <- contr.treatment(levels(rehab$treatment))
contrasts(rehab$treatment)
```

`contr.treatment()` creates the dummy-coding matrix. Assigning it with `contrasts()` tells R to use this coding when fitting a model with the treatment factor. Printing the matrix shows how each treatment is represented.

Effects coding for center

We could also use dummy coding for center. With Center A as the reference, the model intercept would then describe standard care specifically in Center A, and the center coefficients would compare the other centers with Center A.

Instead, we use **effects coding** for center. This describes each center contribution as a deviation from the equally weighted average across centers. The six center contributions sum to zero:

$$c_A + c_B + c_C + c_D + c_E + c_F = 0.$$

Combined with dummy coding for treatment, this gives the intercept a useful interpretation: **the expected standard-care outcome averaged equally across the six centers**. Each center contribution describes how much higher or lower its expected outcome is than that average, comparing patients receiving the same treatment. The treatment coefficient still gives the enhanced-minus-standard difference.

```
contrasts(rehab$center) <- contr.sum(levels(rehab$center))
contrasts(rehab$center)
```

`contr.sum()` creates the effects-coding matrix. Inspect the result: there are six rows, one per center, but only five columns. Centers A–E each have a 1 in their own column and 0 in the others. Center F has −1 in all five columns. Its contribution is therefore minus the sum of the other five contributions; it is not set to zero.

R labels the five columns `1` to `5`, so the fitted center coefficients will be called `center1` to `center5`, corresponding to Centers A–E. Only five center coefficients are needed because the sum-to-zero constraint determines the sixth.

Factor	Coding	Interpretation of its coefficients
Treatment	Dummy coding	Enhanced minus standard care
Center	Effects coding	Each center's deviation from the equally weighted center average

The intercept is a mean for **standard care**, because standard care is coded 0. Averaging over centers does not also average over treatments. In this balanced trial, all centers contribute the same number of patients, so equal weighting of centers also agrees with weighting by their patient counts.

These coding choices change how we express the model, but do not change its fitted outcomes or the estimated treatment difference and its standard error. We set the coding on each factor explicitly, so the analysis does not depend on previously changed global R options.

Fit the fixed-effects model

Specify the model

Let Y_{ij} be the outcome for patient i in center j, and let T_{ij} equal 0 for standard care and 1 for enhanced care. We represent differences between centers through a separate center contribution:

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

With effects coding for center, we impose $\sum_{j=1}^{6} c_j = 0$. The intercept represents the expected outcome under standard care, averaged equally across the six centers. Each c_j is a fixed parameter describing that center's deviation from the average, comparing patients receiving the same treatment. We estimate five center contributions directly and recover the sixth from the sum-to-zero constraint.

The treatment coefficient is the enhanced-minus-standard difference. This model assumes that the expected treatment difference is the same in every center: it contains no treatment-by-center interaction. The remaining errors are assumed to be iid Normal with mean zero and common variance, after accounting for treatment and center.

Connect the model to the design matrix

The equation above describes the center contribution directly. To fit the model, we represent that contribution using the five effects-coded predictors introduced earlier. Written in terms of those predictors, the same model is

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_A Z_{Aj} + c_B Z_{Bj} + c_C Z_{Cj} + c_D Z_{Dj} + c_E Z_{Ej} + \varepsilon_{ij}.$$

Here, $Z_{Aj}, \ldots, Z_{Ej}$ are the numerical values assigned to center j by the coding scheme:

Center	Z_A	Z_B	Z_C	Z_D	Z_E
A	1	0	0	0	0
B	0	1	0	0	0
C	0	0	1	0	0
D	0	0	0	1	0
E	0	0	0	0	1
F	−1	−1	−1	−1	−1

For a patient in Center A, the five center terms reduce to c_A. For a patient in Center F, they reduce to

$$c_F = -c_A - c_B - c_C - c_D - c_E.$$

This is how the coding makes the six center contributions sum to zero. It also explains why the intercept represents the expected standard-care outcome averaged equally across centers.

The **design matrix** collects the numerical predictor values used to fit the model. Each row represents one patient and contains seven entries: a 1 for the intercept, the treatment indicator T_{ij}, and the five center predictors. The center values are shared by all patients in a center, while the treatment indicator depends on the patient's assigned treatment.

Fit the model and inspect the design matrix

```
model_fixed <- lm(distance ~ treatment + center, data = rehab)
summary(model_fixed)
```

In an R model formula, the outcome is on the left of `~`, and predictors are on the right. An intercept is included automatically.

We can inspect the first six patient rows of the actual design matrix used by R:

```
head(model.matrix(model_fixed))
```

`model.matrix()` extracts the design matrix from the fitted model, and `head()` displays its first six rows. The columns are called `(Intercept)`, `treatmentEnhanced`, and `center1` to `center5`. They correspond to 1, T_{ij}, and $Z_{Aj}, \ldots, Z_{Ej}$ in the coded equation.

The `+ center` term adds the five effects-coded columns to the model. For example, `center1` estimates the contribution for Center A: a positive value indicates a higher expected outcome than the center average, for either treatment. The model-predicted standard-care outcome in Center A is the intercept plus `center1`.

> Question 3
>
> Interpret the intercept, the treatment coefficient, and `center1`. How would you recover the contribution for Center F? Why would the intercept have a different interpretation if we used dummy coding for center?

Assess the model results

Treatment effect and uncertainty

In `summary(model_fixed)`, the row `treatmentEnhanced` gives the estimated enhanced-minus-standard difference, its standard error, and a test against a zero difference. The residual standard error at the bottom of the output estimates the individual-error standard deviation after accounting for treatment and center.

Use `confint()` to obtain the 95% confidence interval for the treatment coefficient, and `sigma()` to extract the residual standard deviation.

```
confint(model_fixed, "treatmentEnhanced")
sigma(model_fixed)
```

The standard error of the treatment coefficient describes uncertainty about the estimated treatment difference. The residual standard deviation describes the remaining variation in individual walking distances. Both are expressed in meters.

> Question 4
>
> What are the estimated enhanced-minus-standard difference, its standard error, and its 95% confidence interval? What is the residual standard deviation, and how does its interpretation differ from that of the treatment coefficient's standard error?

Overall contribution of center

The coefficient table describes individual center effects. An **analysis of variance (ANOVA) table** also provides an overall test of whether expected walking distance differs between centers after accounting for treatment.

```
anova(model_fixed)
```

`anova()` produces a row for treatment, a row for center, and a row for the residual variation. With treatment followed by center in our model formula, the center row assesses the contribution of center after treatment has been included.

- `Df` gives the degrees of freedom. Center has five because six center effects are linked by the sum-to-zero constraint.
- `Sum Sq` gives the variation attributed to each term, expressed as a sum of squares.
- `Mean Sq` divides each sum of squares by its degrees of freedom. The residual mean square estimates the individual-error variance.
- `F value` compares a term's mean square with the residual mean square.
- `Pr(>F)` is the p-value for the corresponding overall test.

For center, the null hypothesis is that expected walking distance is the same in all six centers for patients receiving the same treatment. Under effects coding, this means that all six center effects are zero. The plot and estimated center effects show the size of the differences in meters; the ANOVA table assesses the evidence for those differences.

> Question 5
>
> What does the center row tell you about differences between centers after accounting for treatment? Relate this result to the pattern in your exploratory plot.

Check the model assumptions

As in ordinary linear regression, residual plots help assess whether the error distribution and variance assumptions are plausible.

```
plot(model_fixed, which = 1)  # Residuals versus fitted values
plot(model_fixed, which = 2)  # Normal Q-Q plot
```

The first plot can reveal changes in residual spread or patterns left in the mean. The second compares the residual distribution with a Normal distribution. Residuals are calculated from the fitted model; they are estimates of the unobserved errors.

> Question 6
>
> Do these plots suggest clear violations of the Normality or constant-variance assumptions? Describe the patterns that support your answer.